Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Digital Product Passports for Added Value in Robotics and Manufacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Other Robotics/Automation Machinery Technologies	Added Value / After-Sales Service	Product–Service Maintenance / Operation / After-Sales Management
Concept of Proposed Technology	Korean SMEs have a unique opportunity to accelerate digitalization and secure future access to the EU market—two strategic goals achieved in one project. With the introduction of the Ecodesign for Sustainable Products Regulation (ESPR), the EU aims to leverage digital data management to improve efficient use and recycling of critical materials throughout product lifecycles. The ESPR introduces so-called Digital Product Passports (DPPs) and corresponding digital infrastructure as an important innovation for collecting information about materials and products at product, batch, and category level. Robotics and manufacturing stands out as a sector where these digital technologies can improve component and material reuse and lead to more environmentally friendly operations.			

	For Korean industry, this is a chance to lead in sustainable, data-driven supply chains. For Korean exporters, some components and equipment will need to live up to stricter data compliance checks in order for exporters to have continued access to the EU market. At the same time, DPPs are a driver for further digitalization, offering smooth data exchange throughout the supply chain, more sustainable product lifecycles and component reuse, and improved awareness of supply chain risks. This project will help SMEs not only meet EU requirements but also leverage DPPs to unlock new business value and strengthen competitiveness. A full and optimal integration of DPP infrastructure entails several challenges. Firstly, a requirements analysis is needed to determine what data is to be collected and shared . Gathering the requirements will involve both legal and process analysis, for aspects pertaining to any part of the supply chain and product lifecycle. A second step is to determine the interactions where DPP data exchange adds value. This involves identifying the relevant stakeholders, and validating use case descriptions for the interactions. It can also involve the design of data-driven product diagnostics. Finally, with data requirements and stakeholder interactions mapped out, the project will move to implementation. This involves implementing a data sharing solution for the supply chain – enabling standardized data exchange, proper access control, and data sovereignty. Additionally, it will involve a study of opportunities for collecting data at the source. TNO is ideally positioned to meet these challenges as a partner to the Korean robotics and manufacturing industry. TNO brings proven expertise thanks to its involvement in EU research projects rolling out the circular economy agenda. Among others, this includes leadership of the CirPass2 project (reference 1), which produced the comprehensive DPP User Stories document and a reference architecture to clarify the workings of DPP digital infrastructure (5,6). In addition, TNO authored a methodology on developing DPP data standards at the product level for the European Commission's Joint Research Centre. Particular to using DPPs for robotics and remanufacturing, TNO is a partner in the ROB4GREEN project; see (8) for a use case report on tire retreading. Finally, for the Dutch Center of Excellence on Digital Product Passports, TNO has maintained the DPP landscape scan (2) and is supporting industrial companies to adopt DPPs into their business. Based on a DPP ecosystem framework, the landscape scan maps out the roles and activities that organizations perform, making it easier for SMEs to find collaboration partners.
Rationale for Joint R&D Project	This RFP combines the operational context of Korean SMEs with the legal context of the EU. Both are essential to address cross-border compliance for international trade. The project will help Korean SMEs to integrate into global sustainable supply chains. On top of legal analysis, TNO brings valuable expertise in designing DPP solutions that fit the stakeholder landscape. TNO is eager to see the application and knowledge transfer of this expertise with Korean SMEs, which offers a chance to study the impact of local processes, interoperability, and governance models. Furthermore, the project will enable both Korean and EU markets and customers to benefit from a more circular economy, where products are more sustainably and transparently produced, and materials are used more efficiently.
Key Words	1. Digital Product Passport 2. Data Space 3. Supply Chain Traceability 4. Data Interoperability 5. Circular Economy

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 6	End of the Project	Level 8
R&D Objectives	- **Identifying domain data requirements** There are several sources of legal requirements on information that must be provided to oversight authorities or that must be available for parties down the supply chain. This includes the general ESPR, and might include further relevant legislation both from EU or Korean jurisdictions. In addition, the project will need to establish how these requirements relate to the operational practice of the partner SME. This could uncover additional data parameters that are worth including in the digitalization process for the sake of added value. Alternatively, it might lead to the relaxing of some data requirements if some parts of legislation turn out not to apply. - **Identifying stakeholder landscape** In order to identify opportunities for collaboration, it is necessary to map out the stakeholder situation in the environment of the SME. Inspired by the framework TNO used for the Dutch DPP landscape scan[1], this work will entail understanding the roles, influence, and DPP-readiness of relevant organizations in Korean industry, including supply chain partners, financial or administrative service providers, and oversight authorities. - **Identifying domain stakeholder situations and data flows** The next challenge is to clarify how data flows between stakeholders. This includes identifying which stakeholders create new data that is shared in the SME's supply chain. It also includes clarifying data access needs, which will be relevant when implementing data access control. Finally, this involves specifying a suitable governance structure for a data sharing solution, as well as identifying the relevant partners which should be on board for such endeavors. - **Defining use cases for validation** From the overview of relevant stakeholders and data flows, it becomes possible to select the most relevant interactions, which can be realized as use cases in the project to validate the data specifications and the data sharing solution. This selection should happen in consultation with relevant stakeholders, identifying the most relevant local challenges and the opportunities with the highest business value potential for SMEs. An additional factor for consideration is that the selection of use cases covers a sufficiently broad range of operations and data requirements to ensure reliable validation of the whole solution. - **Implementing data sharing solution, demonstrate use cases** To demonstrate that the goals of the use cases are attainable, the project will implement a data sharing solution that allows a realistic simulation of the use cases that were defined. This will include an investigation of suitable data sharing standards and suitable governance models for the group of stakeholders involved in the data sharing activity. Using Validating on the use cases will require a combination of all the outcomes mentioned so far: accurate and complete data models, identification of relevant stakeholders, a precise use case script to simulate using the data sharing solution, and the ability to explain the value that is being demonstrated that way.			
R&D Target Specifications	- **DPP data model aligned with EU regulation and Korean SME needs** The data model should reflect at minimum the requirements of the ESPR and applicable product-specific legislation. It might reflect other EU or Korean regulations as well. The data model should be described in such a way that it allows for the inclusion of domain and metadata semantics, to make the data model more interoperable, so that it can potentially be adopted by a large number of stakeholders. - **Stakeholder map for SME supply chain and ecosystem**			

[1] https://coe-dsc.nl/knowledge-base/community/digital-product-passport-initiatives/

	The stakeholder map will identify relevant partners for collaboration. Each stakeholder will be characterized in terms of a description, the type of organization, and the relevant activities or services it provides. - **Architecture for secure, standardized data sharing** The architecture will enable standardized data sharing that can be deployed by a stakeholder or their service provider. The data sharing solution will support *data sovereignty* and data access control, so that stakeholders stay in control of what data is shared for what purposes. - **Use case validation based on SME challenges and business value opportunities.** The project will identify around 3-5 use cases that represent potential for business value for the SME. Each use case will be defined by a detailed script, identifying the sequence in which stakeholders exchange data, as well as the information contained in the data that is needed to realize the added value of the data sharing. The validation will be performed by following the script and simulating each step in the data sharing solution, to demonstrate the technical feasibility of the use case.
R&D Plan	- **Pilot phase** Time period: Months 1-6 The main activity for the pilot is to identify opportunities for the R&D project. By the end of the pilot the project will deliver an intermediate deliverable of Work Package 1 reflecting requirements and stakeholder analysis, as well as an indication of the topics to be addressed by Work Packages 2 & 4 on the technical part of the data sharing. - **Work Package 1 – Requirements & Stakeholder Analysis** Time period: Months 1-12 Task: Legal analysis, stakeholder analysis Milestone: DPP data requirement mapping, landscape scan of SME's supply chain stakeholders. - **Work Package 2 – Technical Architecture & Governance Design** Time period: Months 7-24 Task: Develop data sharing architecture based on the state of the art in data ecosystems research. Design data sharing governance model that fits with the stakeholder analysis performed in Work Package 1. Milestone: Deliver blueprint for secure data exchange infrastructure and governance model. Intermediate version after month 12 to support Work Package 4. - **Work Package 3 – Stakeholder Engagement & Use Case Development** Time period: Months 13-24 Task: Conduct workshops to identify local challenges and opportunities. Milestone: Report defining 3–5 SME-relevant use cases. - **Work Package 4 – Data Sharing Implementation** Time period: 13-24 Task: Software development Milestone: Software repository for a data sharing node to set up data sharing networks during the use case demonstrations. Intermediate version will be published after month 18 to support Work Package 5. - **Work Package 5 – Use Case Validation** Time period: Months 19-30 Task: Software development, data modeling, presenting results Milestone: Report describing how use case goals were met, per use case. Code repository containing relevant configurations and data models per use case. - **Work Package 6 – Finalization & Follow-ups** Time period: Months 28-36

	Task: Writing, organizing Milestone: A general project report summarizing outcomes. Where needed: publication of training materials to re-apply the method. Where needed: organize public events to encourage further collaboration in the supply chain.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- **Requirements Modeling** TNO will bring expertise in legal analysis for data requirements, in particular for regulations relating to DPPs and other import control or supply chain traceability regulations. TNO will also bring expertise on how to reflect legal insights and provenance in digital applications, enabling robust compliance even in the face of regulatory change. - **Technical Architecture & Stakeholder Analysis** TNO will bring expertise in system architecture, notably as demonstrated in the CirPass 2 project (reference 6). For stakeholder analysis and governance design, TNO will also bring in experience on collaborative business modeling - **Data Sharing Infrastructure Implementation** TNO will bring expertise in software development, particularly on the topic of data sharing. For this, TNO can lean on extensive experience as an active contributor to data space standardization efforts, for example to the Dataspace Protocol (9).
Expected R&R of Korean SME Partner	- **Legal analysis of data requirements** The SME will provide domain expertise for the products or components about which data is legally required. This includes knowledge about existing data modelling efforts which the project might choose to align with. The SME's domain expertise will also be relevant for identifying data fields that are worth including because they add value in the value chain. Finally, where relevant, the SME will provide legal expertise to study Korean legal sources that add further data requirements to the project. - **Stakeholder analysis & use case design** Domain experts at the SME will support TNO to adapt and apply the stakeholder analysis methodology to the local operational context. In particular, the SME will provide knowledge and contacts of the local stakeholder and supply chain situation and help to organize surveys and other data gathering methods. During the use case design, the SME will be key in evaluating the relevance and potential value add, in order to select the most relevant use cases for further development and validation. - **Data capture** It is possible that the project identifies requirements for data which is not yet being adequately captured or provided. Where needed, the SME will provide technical expertise to study the possibilities for improving this data provision in the SME's operational context. This might include knowledge on measuring equipment status or tracking process steps, as well as IT knowledge on storing, modeling and transfering this data, and including relevant metadata. - **Adoption & integration** The SME will provide IT expertise to advise the project on the SME's needs for adopting and integrating any solutions developed in the project. This will be particularly relevant during the work of Work Package 2 on the technical architecture design, and Work Package 4 on the implementation of the architecture. - **Validation** Throughout the project, the input of the SME will be needed to validate the deliverables of the project, again particularly with an eye on validating that it fits the local and operational context and addresses relevant business challenges. As stated, the SME will advise in selecting and developing the most relevant use cases. The SME's knowledge of the local context will be needed to evaluate the success of the use cases' implementation.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	- **Overview of data requirements for compliance and value creation** The project will provide an inventory of what data is to be tracked and passed along the supply chain in order to both comply with regulations and increase the value of products and materials. This outcome also involves legal analysis to clarify the provenance of specific requirements. In addition, it will involve a detailed technical proposal for standardizing the data model. This knowledge will help the Korean SME to pursue ambitious sustainability and circular economy goals. - **Knowledge of supply chain analysis for leveraging DPPs** The project will identify relevant stakeholders in the SME's supply chains to collaborate with on the topic of DPPs. This can include both improving the quality or scope of data collected by the DPP, as well as leveraging the data for additional value creation. In the course of performing this landscape analysis, TNO will also facilitate knowledge transfer to the SME, so that these analyses can be repeated when supply chains change or when other circular economy initiatives ask for similar insights. - **Detailed application of the solution to specific problems (use cases)** Since the project validates its DPP solution on a use case basis, there will be detailed, technical insights into the workings of the DPP data sharing solution and the proposed data model. For an example of this, see the newspost and report in references (3-4) on a vehicle battery DPP demo. Through the use cases the project provides the foundation for expanding the scope of application to involve new stakeholders, processes, or products. Overall, this readies the Korean SME for a high level of digitization, improving its competitiveness.

6. References

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Huib van Nieuwenhuijze is a Senior Researcher at TNO's Data Ecosystems department and is a leading software architect in multiple DPP programmes. His work centers on designing and aligning Digital Product Passport (DPP) system architectures with European regulatory frameworks such as the ESPR, and translating regulatory requirements into technically implementable solutions.

He plays a key role in European initiatives including CIRPASS2, and work with Dutch Customs where he contributes to defining trust and interoperability mechanisms, and shaping cross-sector DPP governance. His expertise spans data interoperability, distributed architectures, standardization, and the integration of DPP concepts into broader digital ecosystems.

Huib is recognized for his structured approach, depth of technical insight, and ability to bridge policy, architecture, and implementation across diverse consortia.

Sjoerd Rongen leads the DPP topic in TNO and is coordinator of the Centre of Excellence for Digital Product Passports (CoE-DPP). Sjoerd plays a key role in national and EU-level initiatives, helping industries prepare for upcoming regulations as well as enabling added value DPP applications by developing interoperable, scalable DPP solutions across sectors like batteries, construction, and textiles.

Dr. Laura van den Aarssen is an innovation strategist at TNO Vector, with over 12 years of experience in enabling value creation and adoption of digital innovations and data sharing across complex ecosystems. Her expertise lies in system thinking, collaborative business modelling, and designing stakeholder-driven innovation trajectories.

Laura has led national and European projects, including serving as lead for the assessment methodology in the EU project Change2Twin (digital twinning in manufacturing). She works across public sectors — including ministries such as EZ, BZK, LNV, and RVO — and private sectors, supporting SMEs, large tech firms (e.g. BMW, Audi, VW), agrifood companies like FrieslandCampina and Agrifirm, and industry associations such as FME, Techniek NL, and the furniture and interior construction sector.

At TNO Vector, she helps organizations navigate uncertainty, foster collaboration, and take meaningful steps toward systemic change — including the adoption of Digital Product Passports (DPPs).

Dr. Kartik Chawla completed his B.A.-LL.B. (Hons). at the NALSAR University of Law in 2017, and went on to pursue his Masters degree with the European Masters in Law and Economics program at Ghent, Hamburg and Rotterdam Universities in 2018. With a PhD in Information Systems and Law from Tilburg University, he specializes in Responsible Innovation, crafting impact-oriented digital architectures that solve real-world regulatory challenges.

After joining the TNO Data Ecosystems department, Kartik was involved in European and national research projects relating to DPP and supply chain digitalization. Kartik contributed legal analysis and data modeling methodology for the ESPR and a variety of relevant domains including batteries, steel, and textiles.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization